

EnableController: Generative Kinematics for Universal Access and Gamified Rehabilitation

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Figure 1: Playing Silksong with EnableController.

Abstract

For millions of people with disabilities, standard game controllers act as physical barriers, and while motion-based rehabilitation exists, it is often restricted to unengaging, custom-made software. The primary goal of this research is to solve a fundamental problem in accessibility where controlling a video game using body motion remains too difficult to configure and too limited in application. Our previous solution, the System for Unified Kinematic Input (SUKI+), successfully simplified the development of these motion controls by normalizing sensor data. However, a significant gap remained for the end-user, who still faced complex technical setups and a lack of compatible games. EnableController bridges this gap by introducing a Generative AI layer to simplify configuration and a universal Emulation layer to expand compatibility. By enabling users to configure complex control schemas through natural language and mapping those inputs to standard virtual drivers, EnableController transforms the human body into a universal controller, making accessibility configurable through natural language and compatible with the broad library of PC games that accept standard keyboard or gamepad input.

CCS Concepts

- **Human-centered computing** → HCI design and evaluation methods; • **Computing methodologies** → Artificial intelligence; • **Applied computing** → Health care information systems.



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Keywords

Generative AI, Accessibility, Rehabilitation, Kinematics, Large Language Models, Game Controllers

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1 Introduction

Video games have evolved from simple entertainment into a primary form of social connection and cultural participation. However, the standard interfaces used to access this world—gamepads, keyboards, and mice—enforce a normative physical dexterity that effectively excludes users with limited mobility or motor impairments. For physical therapy patients, this exclusion is compounded by the nature of rehabilitation itself. Therapeutic exercises are often repetitive, unengaging, and disconnected from immediate feedback, resulting in low adherence, while body-tracking games for the Kinect, Wii, or VR are not customizable to meet the limitations of individuals with impairments. Patients desire the agency to play popular titles like *Minecraft* or racing simulators, yet they are frequently forced to use rigid, clinical software because standard games cannot interpret their therapeutic movements. To solve this, we need a system that can translate any physical ability a user possesses into the digital inputs required by modern software.

2 The Challenge and the Foundation

Existing solutions typically fail because they are either too rigid or too complex. Most motion-control systems function as walled

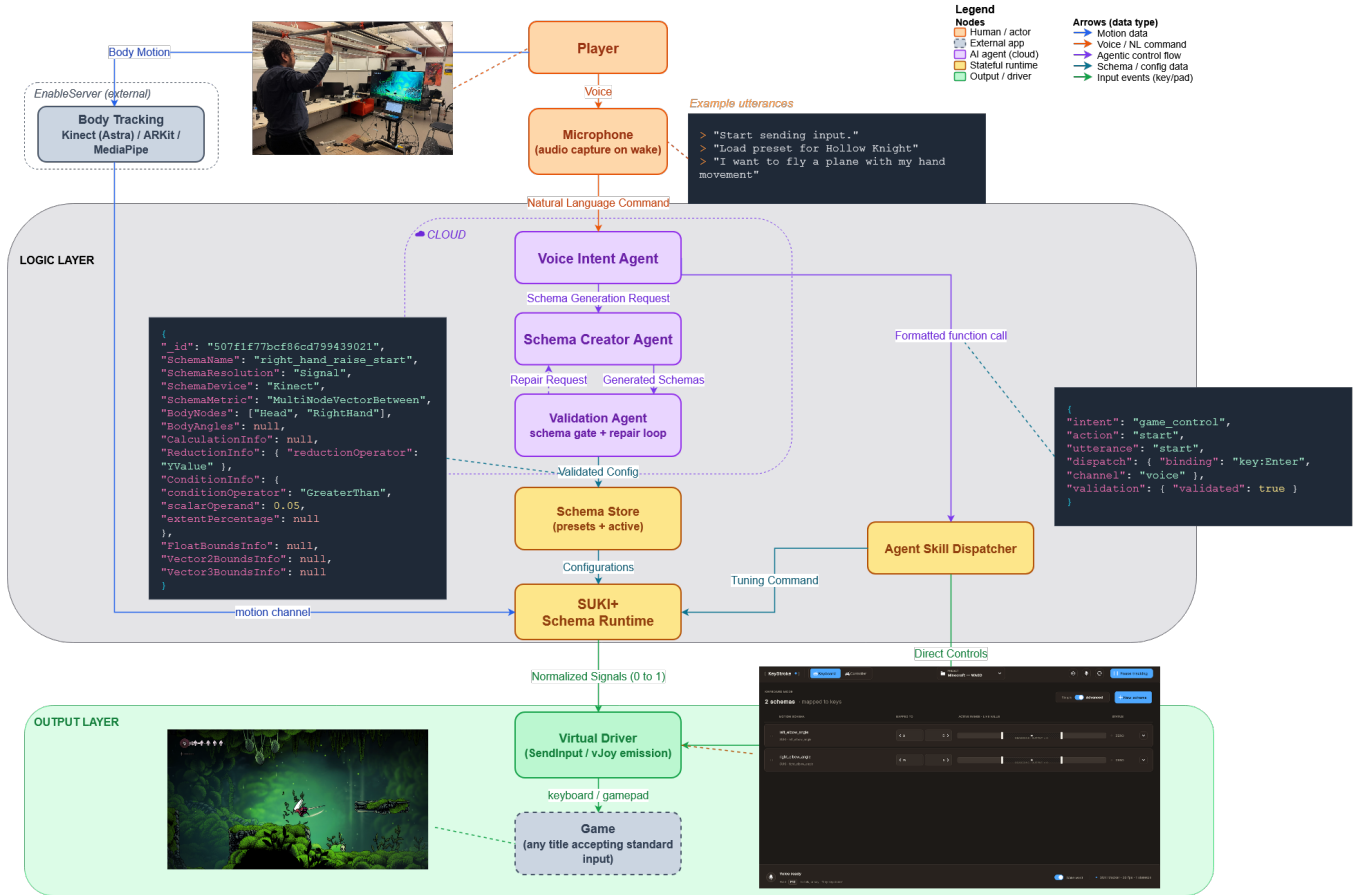


Figure 2: EnableController system architecture. Body motion captured by Kinect, ARKit on iOS devices, or webcam pose estimation is normalized and mapped through SUKI+’s active schema configurations, then emitted as keyboard, mouse, or virtual Xbox controller signals via OS-level driver injection. Schema configurations come from two sources: a curated preset library loaded at session start, and the Gemini-powered LLM schema generator invoked on demand through natural-language voice commands. Voice control also tunes active configurations directly during use.

gardens, restricting users to the specific games that ship with the hardware. Furthermore, tools that do allow for custom mapping often present a steep technical barrier, requiring users to manually adjust sensitivity sliders, dead zones, and axis thresholds. Our previous work, the SUKI+ framework, addressed the hardware side of this problem by developing a universal translator for body-tracking solutions that supports expanded biometric sensors and custom robotic inputs [Shang et al. 2025]. SUKI+ allowed developers to build motion applications without worrying about body tracking at all, while readily supporting various body-tracking technologies, such as a Kinect, an iPhone, a custom tracker, or even a standard webcam. While this was a breakthrough for developers, it did not eliminate the friction for end users, who still struggled with complex configuration menus and a limited selection of compatible games.

3 The Solution: Generative Configuration

enAbleController solves the configuration problem by integrating Large Language Models directly into the control loop. We have replaced the complex graphical interfaces of traditional mapping software with a conversational agent powered by Gemini. Instead of navigating drop-down menus to assign a joint to an action, a user simply describes their intent in plain English. A command such as “allow me to steer by tilting my head” is parsed by the AI, which infers the intended joint and motion type from the description and generates a corresponding schema for review. This allows a therapist to modify a patient’s range-of-motion requirements just by speaking, or a gamer to tweak their control sensitivity mid-game without ever needing to write code or understand vector mathematics.

Because the LLM’s output is not always correct, EnableController applies multiple layers of defense before a generated configuration reaches the user. A rule-based schema validator inspects every

generated schema for structural correctness including name format, locked enum values, allowed body joints and angles, bounds sanity etc. and rejects malformed output before it is saved. When validation fails, the system performs a single multi-turn repair attempt, returning the validation errors to the LLM and requesting a corrected schema. At the user level, an explicit voice undo command, a five-step undo history scoped to the active session, and a safe-default reset to the loaded preset together allow users to experiment freely with conversational reconfiguration. The preset library shown in Figure 2 also serves as a stable fallback when generative configuration is undesired.

EnableController dispatches voice commands through an agentic skill router rather than a single monolithic prompt. Each command verb like pause, mode switch, preset load, schema generation, schema modification, undo is implemented as a self-contained skill with its own parameter schema and handler. The LLM's role at dispatch time is to select among these skills via native function calling, with each skill carrying a description that disambiguates it from overlapping commands. This factoring scales the command vocabulary independently of prompt length and isolates the safeguards (validator, repair retry, undo history) within the skills that need them, rather than entangling them in shared dispatch logic.

4 The Expansion: Universal Emulation

EnableController solves the compatibility problem by adding a Virtual Emulation Layer that fundamentally changes how the operating system perceives the user. By using virtual driver injection, the system maps the user's body movements to the signals from a standard Xbox controller, mouse, or keyboard. To the operating system, EnableController appears as a standard gamepad, allowing input from the user's body to drive any game that accepts gamepad input, whether from the user's elbow, head, or smile. This unlocks immediate compatibility with a substantial portion of the existing PC game library. A stroke survivor can now use their prescribed arm extension exercises to cast spells in an RPG, allowing prescribed therapeutic motions to drive the same commercial games that non-disabled players enjoy.

5 Implementation and Evaluation

EnableController is implemented in Unity 6 and runs on Windows. The Generative AI layer is provider-agnostic and currently supports both Google Gemini and Anthropic Claude through unified function-calling abstractions; reported numbers below use Gemini 2.0 Flash, selected for its low per-call cost in deployment. The Emulation layer uses InputSimulator for keyboard and mouse output and Nefarius.ViGEm.Client (with the ViGEmBus driver) for virtual Xbox controller injection. Body tracking is supported via Kinect, ARKit on iOS devices, and webcam-based pose estimation through the SUKI+ framework, allowing the same configuration to work across heterogeneous hardware.

We conducted internal evaluation of the system's most uncertain component: LLM-driven schema generation. Across 50 natural-language prompts spanning arm, leg, head, torso, and full-body motions, 92% produced valid schemas on first generation. The rule-based validator rejected the remaining outputs as malformed; of these, 87.5% were successfully recovered by a single multi-turn

repair retry, yielding an end-to-end success rate of 98%. End-to-end motion-to-input latency, measured at the schema tick boundary across our three demonstration titles, varied modestly across tracking modalities. Kinect delivered the lowest latency (median 40 ms, 70 ms), followed by ARKit on iOS (77 ms / 110 ms) and webcam-based pose estimation (124 ms / 224 ms). All three modalities operated well below the threshold for real-time game input. Voice-command latency averages approximately 3.5–4 seconds end-to-end, an acceptable cost for setup-phase reconfiguration that does not require frame-accurate response.

The booth installation features three commercial PC games, including a platformer (Hollow Knight), a driving and survival title (Pacific Drive), and a puzzle game (Portal 2), selected to span distinct input requirements. Attendees choose a tracking sensor (Kinect or iOS-based ARKit), select a game from the menu, and play short sessions of three to five minutes. Voice commands let attendees reconfigure mappings live, demonstrating the conversational configuration loop. Both seated and standing modes are supported, reinforcing the project's accessibility framing.

6 Limitations and Scope

EnableController is designed for users who retain useful gross motor capability, including those building motor capability through rehabilitation. It complements rather than replaces subtle-input accessibility systems such as gaze tracking and facial-gesture recognition, which serve users whose primary capabilities lie outside the gross-motor domain. LLM-driven schema generation produces correct configurations at high but imperfect accuracy; errors are caught structurally rather than relying on user vigilance. Formal user studies with disabled populations beyond our existing rehabilitation context are planned but not yet conducted, and represent the natural next step toward a full HCI evaluation.

7 Conclusion

The future of accessibility lies not in building specialized games for specific needs, but in building tools that allow everyone to play the same games. EnableController reduces the technical and physical barriers between the human body and the digital world by combining natural-language configuration with universal controller emulation. We empower users to define their own way to play, ensuring that technology adapts to the human rather than forcing the human to adapt to the technology.

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